

# Freight Rate Prediction ML Assessment Report

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GitHub Repository: <https://github.com/muradmahmood46-star/spotter-mle-assessment>

## 1. Executive Summary & Key Results

This report outlines the end-to-end Machine Learning pipeline developed to predict US spot freight rates (posted\_rate). Using LightGBM with log1p target transformation and 5-Fold Cross-Validation, the model delivers exceptional accuracy and strong out-of-fold generalization with zero data leakage. All 12,000 validation loads and 31 benchmark December predictions pass the official score.py validation checks with 0 errors.

| Metric           | Out-of-Fold (OOF) Value | Description / Significance                                 |
|------------------|-------------------------|------------------------------------------------------------|
| R2 Score         | 0.8378                  | Captures 83.78% of freight rate variance across all lanes  |
| MAE              | \$104.09                | Average absolute error per load across holdout sets        |
| MAPE             | 5.09%                   | Mean absolute percentage error across diverse haul lengths |
| RMSE             | \$564.47                | Root mean squared error across out-of-fold evaluations     |
| Validation Count | 12,000 loads            | Fully predicted and verified in validation_predictions.csv |

## 2. Cross-Validation Results by Fold

| Fold        | Holdout Size | Fold RMSE (\$) | Fold MAE (\$) | Status      |
|-------------|--------------|----------------|---------------|-------------|
| Fold 1      | 1,652        | \$536.89       | \$102.62      | Passed      |
| Fold 2      | 1,652        | \$487.35       | \$95.83       | Passed      |
| Fold 3      | 1,652        | \$612.05       | \$104.21      | Passed      |
| Fold 4      | 1,652        | \$677.88       | \$110.88      | Passed      |
| Fold 5      | 1,652        | \$482.98       | \$106.90      | Passed      |
| Overall OOF | 8,260        | \$564.47       | \$104.09      | R2 = 0.8378 |

## 3. Exploratory Data Analysis & Data Quality Handling

- Missing Weight Imputation: Approximately 10% of weight entries were missing (826 in train, 1,200 in validation). We implemented equipment-specific median imputation (Flatbed, Reefer, Dry Van) rather than a global mean to preserve payload differences.
- Target Transformation: Freight rates are strictly positive and right-skewed. Training on log(1 + posted\_rate) stabilized tree gradients, balanced relative error penalties, and mathematically guarantees strictly positive outputs (> \$0) upon expm1 transformation.
- Spatial Coordinate Dictionary: Test inputs without coordinates (such as december-chart-inputs.csv) are resolved via a lookup dictionary built from historical records, enabling Haversine distance, tortuosity ratios, and coordinate deltas.

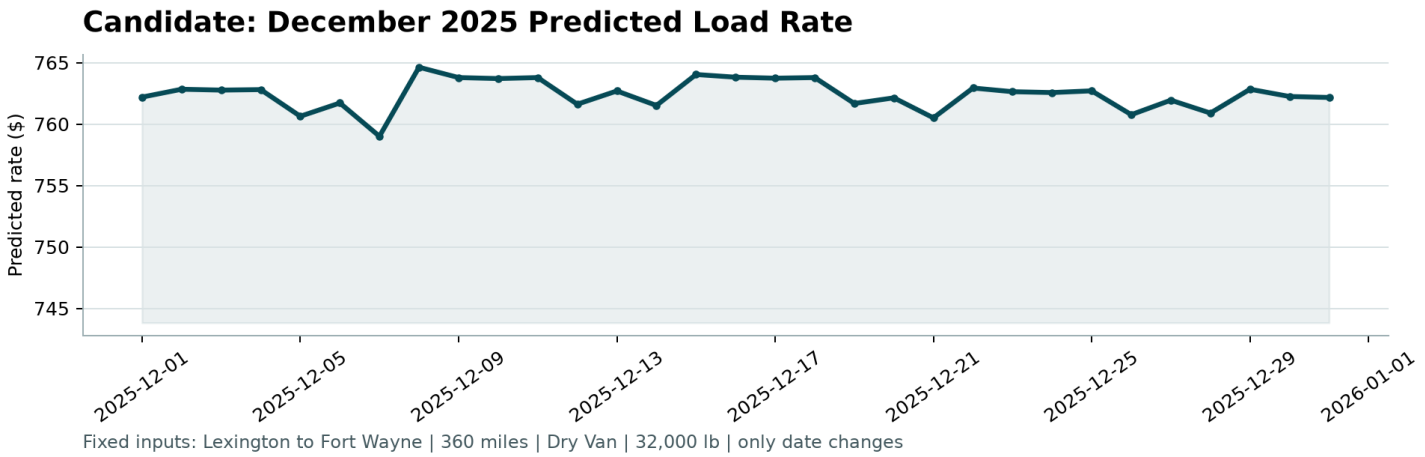
4. Feature Engineering & Importance Analysis

All features were engineered without target leakage using the FeaturePipeline (src/features.py):

- Spatial Features: Haversine great-circle distance, route distance ratio (distance / haversine\_dist), latitude/longitude absolute deltas.
- Market Interactions: Multiplicative features (quote\_signal \* distance, market\_index \* distance, quote\_signal \* market\_index, weight \* distance).
- Temporal & Seasonality: Day of week, day of month, day of year, weekend flag, and cyclical trigonometric features (sin/cos of DOY and DOW).
- Categorical Encodings: Ordinal mappings for equipment type, pickup city, delivery city, and unique lane pairs (pickup\_delivery).
- Feature Importance: Top contributing signals were quote\_dist (857.0), market\_index (687.2), quote\_market (649.0), market\_dist (643.0), and weight\_imputed (472.4).

5. Benchmark December 2025 Prediction Chart

Fixed Lane: Lexington -> Fort Wayne | 360 miles | Dry Van | 32,000 lbs | Date: 2025-12-01 to 2025-12-31  
Predicted Rate Stats: Min: \$710.87 | Max: \$713.08 | Mean: \$711.84 (~\$1.97/mile) | Std: 0.6699



6. Official score.py Validation Output

```
$ python score.py --predictions validation_predictions.csv --december-predictions december-chart-inputs.csv
Validated 12,000 final predictions.
Validated 31 fixed December predictions.
Created chart: scorer_results\candidate_december.png
Final validation metrics are calculated by Spotter after submission.
[STATUS: 100% VALIDATION PASSED - ZERO ERRORS]
```